Batches™

Rapid Microneedle Patch Test for The Blistering Virus

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A new widespread virus was discovered, which causes blisters on the hands and feet. Our proposal to combat this new virus is Batches™. Batches™ is a revolutionary patch device that is self-administered by patients to measure blister fluid for trace elements of the virus. Overall, this product is easy-to-use, environmentally friendly, and effective.

Biomarkers chosen for Batches™ are blister fluid proteome, glycoprotein, and Eosinophil Cationic Protein (ECP). The ratio of blister fluid proteome, which is a group of proteins found in proteins, to the blister fluid or serum, is found to be 1:5, at 16 mg/L to 80 mg/L[1]. Glycoprotein is also a protein group found in the serum and increases proportionally to the amount of serum in the blister.[2] Therefore, it will be more advantageous to put this protein group in the spotlight depending on the blister's size. ECP can be related to bullous pemphigoid (BP), an autoimmune blistering disease, as ECP is released in response to BP levels in the blood.[3]

The Batches™ design will be in the form of an adhesive patch that can easily be placed by the general population. The tape is viscoelastic so that it can stay on the user's hand as they are moving about, waterproof, and see through. The patch design is around the size of a typical bandaid for familiarity, easy storage, and quick application. Microneedle patches will not be too invasive and can be likened to nicotine patches, which are already FDA approved. Since Batches™ will not pose that much harm to the user, it is likely a Class I device, which is largely exempt from regulatory process because of how safe it is. This means the pre-clinical trials on humans or animals can be disregarded. As opposed to a syringe, a microneedle patch will be more universal because it will not need to account for differences in skin elasticity, texture, or blister depth and breadth. Furthermore, the patches are environmentally-friendly, as they (with the exception of the microneedles) will be biodegradable. The patch's microneedles will need to be safely disposed of, as it is considered as biohazardous waste. In our product's package, a bag for disposing of biohazardous waste will be included for safe disposal. Finally, the material that comprises Batches™ is cheap enough to be mass produced, so a profit can easily be made to sustain the company.

The microneedle will be 1000 microns in length, 200 microns in width, a few microns in diameter, and an approximate thickness of 1-25 microns to create a transport pathway for the sampling method. Even though 3D printers are able to operate in dimensions of 20 microns, this will be an incredibly expensive endeavor to do en masse. Methods like thermoforming is a cheaper alternative, as it simply involves heating a sheet to mold it to a specific shape and then cooling it down. Polycarbonate, a transparent material, will be the base chosen for the microneedle because of its biocompatibility.

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[1] https://analyticalsciencejournals.onlinelibrary.wiley.com/doi/epdf/10.1002/pmic.200600938

[2]https://reader.elsevier.com/reader/sd/pii/0305417978900050?token=B74F382AE60744ACBF49926DB7653A50C24E5AA02E336A81E2F4B5F83FAC2FF33418CC1FE80A34FD792D81D1B4633310&originRegion=us-east-1&originCreation=20221001182157

[3] https://onlinelibrary.wiley.com/doi/10.1111/jdv.12464

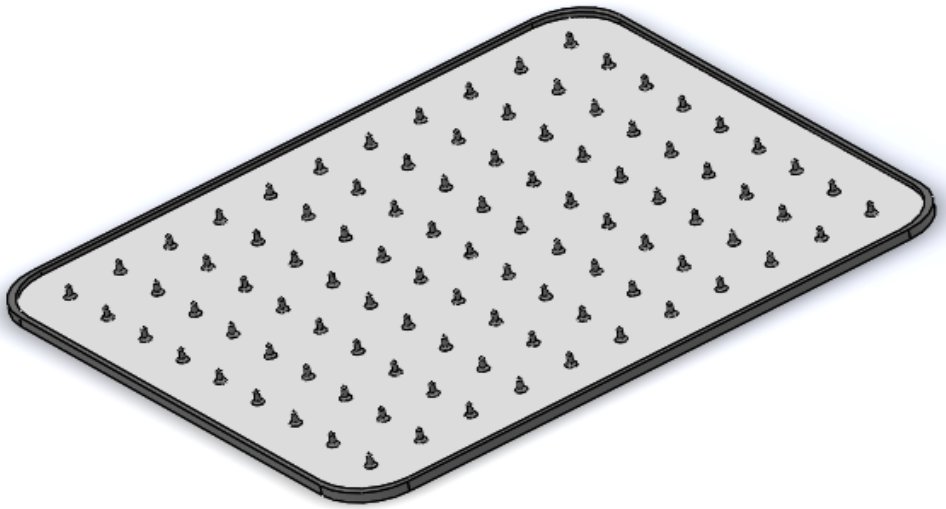

Figure 1: Microneedle patch design.

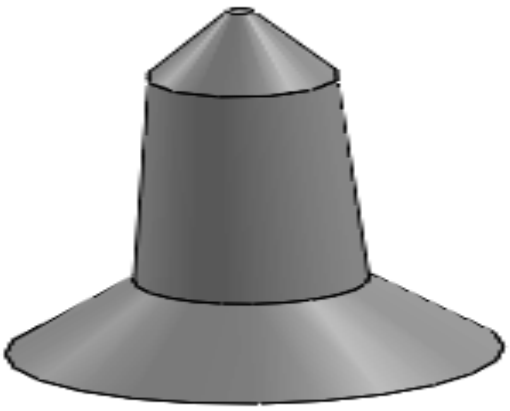

Figure 2: Model of microneedle of the patch.

Inside Batches™ will be a miniature LFA that will operate on the same principle as a rapid antigen device. ELISA, an assay technique used to directly compare soluble substances such as proteins, will be the process used to detect and quantify any of the biomarkers. ELISA will use two antibodies to "sandwich" the protein identifier, but since ELISA is largely used in laboratories, it is likely more prudent to commit to an LFA inside the micropatch itself. However, should you prefer, the test results from the micropatch can be sent to a laboratory for this method for further analysis.

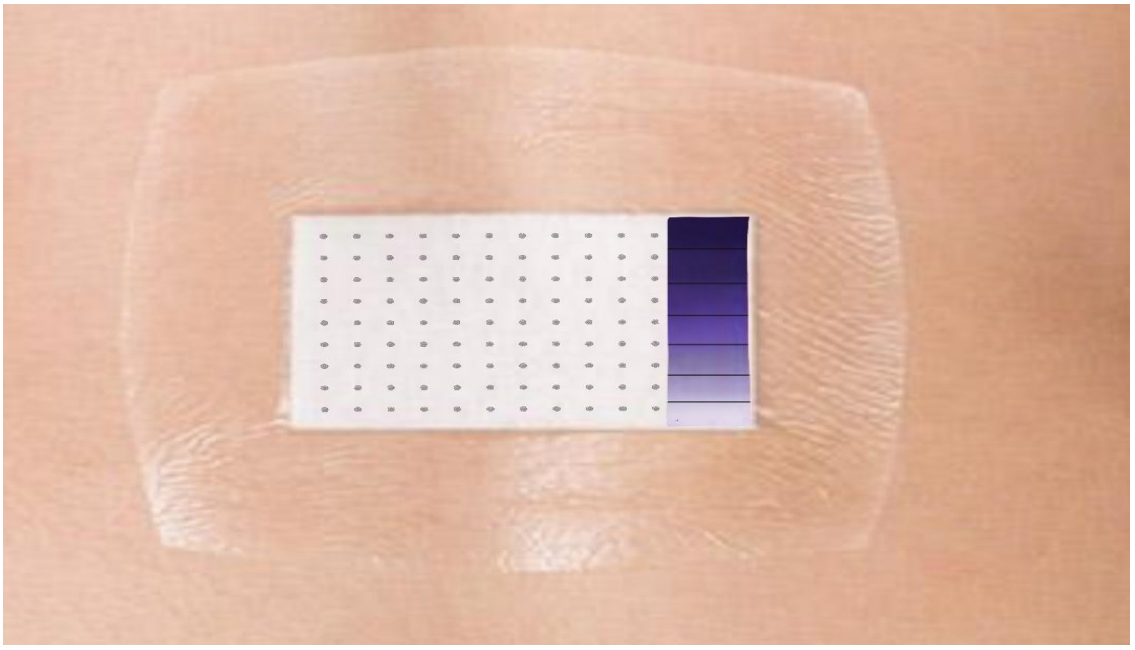

Figure 3: Batches™ on a finger.

If any of the biomarkers are detected on the blister, Batches™ will change in color depending on how prevalent the presence of the biomarkers are. There are two proposals: one proposal on how Batches™ should work was based on COVID-19 rapid antigen testing. The test will involve fluorescent dyes that will activate in the presence of antigen contained by a semipermeable membrane, so the dye can be activated without leaving the patch. There will be three layers that will contain the fluid using hydrogels. The fluorescent dye will be contained by hydrogels, and when it is placed on a blister that contains the virus, protein diffusion will occur, which will then activate the dye.[4] This means that traces of proteins must be left within the fluorescent dye, but not enough to induce a great color change. However, considering that rapid antigen tests were sometimes unreliable in their result by giving false negatives, it will be better to pivot to a more accurate test.

Another proposal for inducing a color change is based on the Biuret Test mechanism, where a solution of copper sulfate and potassium hydroxide is used to detect any protein samples. The solution will change from a blue color to a purple color depending on the proportion of protein in the serum. Since blister serum is partially composed of proteins itself, a person afflicted by the unknown disease should look for a vibrant color change. On one side of the patch, there will be a control group that can be compared with the color of the patch in direct contact with the blister.

[4] https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6324205/

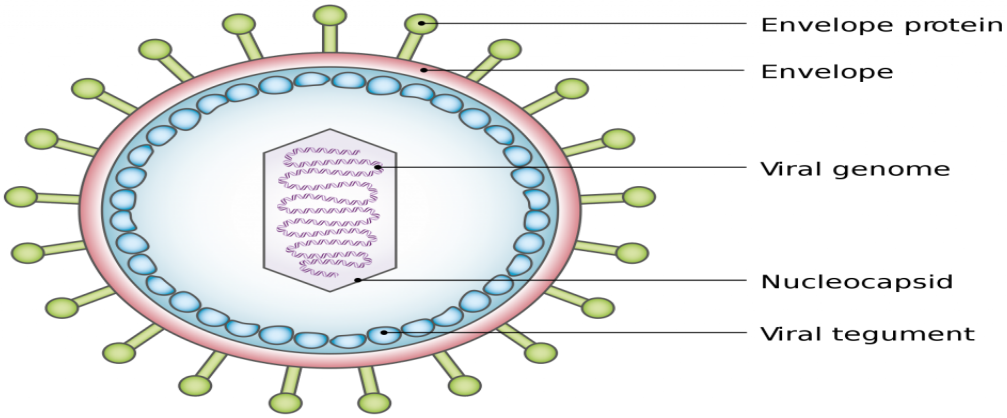

Figure 4: Virus diagram.

For accessibility uses, an app program is developed for the colour analysis of the patch. This will allow for the quantification of how severe the affliction is. It can also be used for people who have color blindness issues. It will also have a text-to-speech function to help with any people who are blind.

The distribution plan for Batches™ will be to supply pharmacies with kits that contain these patches for public distribution. Since it is a widespread disease, it is likely in any government's best interest to subsidize such endeavors to prevent or reduce any endemics. Should any bleeding be induced by the microneedle patch, alcohol wipes and bandaids are recommended.

This concludes the proposal for Batches™. It will be a microneedle patch that is 10 mm by 12 mm by 1 mm composed of polycarbonate material. Batches™ will also house either a semipermeable membrane for the fluorescent dye during antigen testing or contain a biuret solution for detecting polypeptide bonds. These are contingent on what type of virus the blister will be, with the former being more specific, and the latter being more general. There will also be an option to perform an ELISA test by sending your micropatch to an appropriate lab for more accurate testing. Batches™ will be manufactured through thermoforming, which will be a cheap way to mass produce the patches. Government subsidies will also sustain this project because the widespread reach of this virus can make this program likened to the COVID-19 response. Even without government subsidies, the low cost of the project will make this feasible, regardless. It will be supplied to pharmacies, hospitals, or clinics to those who may be afflicted for self-testing.

Batches™ will have a long-lasting impact as an at-home testing kit. As shown by the Covid-19 epidemic, at-home testing is crucial for disease control in a modern society. Especially due to more frequent disease outbreaks because of climate change. Batches™ can help mitigate the outbreak of this bullous virus. Batches™ technology can be reapplied to detect other diseases –that are not limited to a blister-causing virus– in response to future outbreaks. In a growing age of medical advancements, Batches™ will be at the forefront of innovation, bringing in a new era of at-home testing.